

Assemble a Six Ingredient Salad SOP

One episode builds one salad and splits it across two plates. Greens and five toppings go into the mixing bowl in a fixed order, the salad is dressed from two squeeze bottles, tossed with the wooden spoon, and portioned onto two plates with the tongs. All six ingredients have been added, the wooden spoon and tongs are back in their rest, and both bottles are back on their spots when the episode ends.

The fill order never changes. Greens, then topping 1 through topping 5 in tray order.

The tools have fixed jobs and each one lives in a rest between uses. The tongs load and plate. The two bottles dress, one at a time. The wooden spoon tosses. No tool is ever held while another tool is being used, and no tool is ever set down outside its rest.

The table is set up in one of three ways. Only the ingredient containers move, and the two plates take the side they leave free; the mixing bowl, the cutting board, and the bottles are in the same place in all three.

- **Config L:** the ingredient containers are on the left side, and the plates are on the right.
- **Config M:** the ingredient containers are at the back-centre, behind the mixing bowl, and the plates are on the left.
- **Config R:** the ingredient containers are on the right side, and the plates are on the left.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the ingredient containers start on the left (**Config L**), at the back-centre (**Config M**) or on the right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Plate side:** the plates are always on the side the ingredient containers are not — the right in Config L, the left in Config M and R — one above the other.
- **Same-side rule:** the gripper on the ingredient containers' side takes the tongs, loads every ingredient, and plates the salad — the **left gripper** in Config L, the **right gripper** in Config M and R. The gripper on the plate side brings and returns each plate. No arm reaches across the table.
- **Fixed roles:** the **right gripper** dresses with both bottles and tosses with the wooden spoon in every config. The fill order, the tool rests, and every handling rule are the same in all three.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** bimanual

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the ingredient containers in the start zone for this episode's config, both plates on the plate side, the cutting board and utensils at the lower middle, the mixing bowl in the middle, and the two bottles at the far right.
3. Both arms are at home with grippers open.
4. The tabletop is clear of anything but the zones listed below.
5. Nothing obstructs the paths between the ingredient containers, the mixing bowl, the cutting board, and the plates.

Materials checklist

1. Six open ingredient containers sit together in the start zone for this episode's config, none of them overlapping:
 - **Config L:** the left side of the tabletop, immediately left of the mixing bowl
 - **Config M:** the back-centre, behind the mixing bowl
 - **Config R:** the right side of the tabletop, immediately right of the mixing bowl

Within the zone: the cucumber bowl at the back left, the leafy-greens bowl at the back right, the shredded-carrot bowl at the middle left, the broccoli bowl at the middle right, the large square chopped-cabbage container at the front left, and the sliced-red-onion bowl at the front right.

2. The large square container holds enough chopped cabbage for the episode, filled to below its rim so the tongs can close inside it without cabbage spilling over the side.
3. Each of the five round ingredient bowls holds enough of a single ingredient for the episode. Nothing is mixed between bowls and no bowl is filled above its rim.
4. The six containers stay in the pictured two-column arrangement. Their positions are fixed at setup and do not change mid episode.
5. A white cutting board sits below the mixing bowl. Clean, dry metal tongs lie diagonally across its upper half with the gripping end to the left and handles to the right. A clean, dry

wooden spoon lies across its lower half with the spoon bowl to the left and handle to the right.

6. The mixing bowl sits empty and dry on its ring near the middle of the tabletop.
7. Two bottles stand upright at the far right, side by side, each on its own spot. Both spouts point upward.
8. Both plates sit on the plate side — the left in Config M and R, the right in Config L, between the mixing bowl and the bottles — empty and right side up, one above the other and not touching. The upper plate is the first plate and the lower plate is the second plate.

Workspace layout

- **Ingredient zone:** the left side (Config L), the back-centre behind the mixing bowl (Config M), or the right side (Config R) of the tabletop. It holds the six ingredient containers in the pictured two-column arrangement. Nothing is ever set down among them.
- **Utensil rest:** the white cutting board below the mixing bowl. The tongs and wooden spoon lie here.
- **Mixing bowl:** near the middle of the tabletop, reachable by both arms. Every ingredient lands here and the whole toss happens here.
- **Bottles:** two bottles stand upright at the far right.
- **Plates:** on the plate side of the tabletop — the right in Config L, the left in Config M and R — one above the other.

Naming: the big empty bowl in the middle is always called the **mixing bowl**. The six filled containers are always called **ingredient containers**, by their contents. Nothing in this document is just "the bowl".

Both arms begin at home. The full ingredient zone, mixing bowl, cutting board, bottle zone, and both plates remain visible to the environment camera throughout setup.

Vocabulary

- **Start zone:** where the ingredient containers stand at the start of the episode — the left side (**Config L**), the back-centre (**Config M**), or the right side (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Plate side:** the side opposite the start zone, where both plates stand — the right in Config L, the left in Config M and R. The **ingredient-side gripper** is the gripper on the start zone's side and the **plate-side gripper** is the other one; in Config M the right gripper is the ingredient-side gripper.

Arm assignments

- **The ingredient-side gripper owns the tray and the tongs** — the right gripper in Config M and R, the left gripper in Config L. It loads all six ingredients and uses the tongs to plate the salad.
- **The right gripper owns both dressing bottles and the wooden spoon** in every config. It dresses and tosses the salad.
- **The plate-side gripper owns the two plates** — the left gripper in Config M and R, the right gripper in Config L. It brings one plate at a time to the portioning position and returns each filled plate. It does not take the tongs or pick anything off the tray.
- **The wooden spoon tosses the salad.** The right gripper uses it alone for the whole toss.
- **The tongs plate the salad.** The ingredient-side gripper uses them to carry portions to the plate brought forward by the plate-side gripper.
- **Nothing is passed hand to hand.** A tool that changes arms does so by going back to the cutting board first.
- A gripper opens only over the mixing bowl, over a plate, at a stable plate placement, or at a stable placement on the cutting board. Never open a gripper over the tray, over a topping bowl, or over open tabletop.

Handling the tools

Each tool has one job and one home. The cutting board is the home for the tongs and wooden spoon.

- **Tongs:** the ingredient-side gripper takes them from above, across both handles near their top ends. Closing the gripper closes the tongs. The tongs load every ingredient and plate the salad.
- **Bottles:** the right gripper takes one bottle at a time around the body, below the neck and above the base. Squeezing the gripper squeezes the bottle. Each bottle goes back to its spot before the other is taken.
- **Wooden spoon:** the right gripper takes it from above by the top of the handle. The wooden spoon tosses the salad and does nothing else.
- **Plates:** the plate-side gripper takes one plate at a time by the rim, keeps it level, and releases it only after it is stable on the tabletop. The gripper never touches the food-contact surface.
- **A tool goes back to the cutting board before the next tool is picked up.** Never hold two tools at once and never lay a tool down on the tabletop, on the tray, or across the mixing bowl rim.
- **Change jobs through the cutting board, never in the air.** No gripper shifts its hold along a handle while the tool is off its rest.
- Carry every tool level and clear of the rim of anything it passes over.

The fill order

- **Tray order, not reach order.** Greens first, then topping bowl 1, 2, 3, 4, 5 in that order, whatever is in them.
- The greens always go in first. They are the base and every topping lands on top of them.
- **One ingredient at a time.** Never take from two containers on one trip and never combine two toppings in the tongs.
- Use as many loads as needed for the intended portion from each topping bowl. The container does not have to be empty before returning the tongs to the rest.

Handling standard

- Bring the tongs down inside the rim of the mixing bowl before opening them, so nothing falls in from height and bounces back out.
- Bring the tongs down inside the rim of a container before closing them, so nothing is dragged over the side.
- **Never grip, steady, lift, or slide the mixing bowl.** The ring holds it. If the mixing bowl has moved off its ring, stop and reseal it before continuing.
- **Never grip an ingredient directly with a gripper.** Everything that goes in the mixing bowl or onto a plate gets there through the tongs.
- **Never press or pack the salad down.** Nothing in this episode presses on the contents of the mixing bowl or of a plate.
- Spilled leaves or toppings on the tabletop are left where they fall. Do not stop to collect them.

Steps

Steps 1, 2, and 5 depend on the config: the **ingredient-side gripper** takes the tongs, loads, and plates (the left gripper in Config L, the right gripper in Config M and R), and the **plate-side gripper** brings and returns each plate (the right gripper in Config L, the left gripper in Config M and R). Steps 3, 4, and 6 and every other line are the same in all three configs.

Step 1: Load the greens

Goal: the intended portion of greens is in the mixing bowl and the tongs are back on the cutting board.

Look where the ingredient containers are before reaching for the tongs.

- **IF the ingredient containers are on the left (Config L):** the **left gripper** takes the tongs from the cutting board, lifts them clear, and brings them down into the greens tub on the left.
- **IF the ingredient containers are at the back-centre (Config M):** the **right gripper** takes the tongs from the cutting board, lifts them clear, and brings them down into the greens tub

behind the mixing bowl.

- **IF the ingredient containers are on the right (Config R):** the **right gripper** takes the tongs from the cutting board, lifts them clear, and brings them down into the greens tub on the right.

Then, in all three:

- Close the tongs on a load of leaves and lift straight up.
- Carry the tongs level to the mixing bowl, lower them inside the rim, and open them to release the load.
- Repeat the same transfer as many times as needed for the intended portion. The tongs stay in the gripper between loads.
- Carry the tongs back to the cutting board, lower them until stable, and release.

Check: the intended portion of greens sits in the mixing bowl and the tongs are back on the cutting board. If leaves are hanging over the mixing bowl rim, push them in with the tongs before returning them.

Step 2: Load the five toppings

Goal: the intended portion from each of the five topping bowls is in the mixing bowl on top of the greens, and the tongs are back on the cutting board.

Work one topping bowl at a time in tray order: topping bowl 1, then 2, 3, 4, and 5.

Run this cycle once per topping bowl, in full, five times.

- **IF Config L:** the **left gripper** takes the tongs from the cutting board and lifts them clear. **IF Config M or R:** the **right gripper** takes them.
- Bring the tongs down inside the topping bowl, close them on the whole portion, and lift straight up.
- Carry the tongs level to the mixing bowl, lower them inside the rim, and open them to release the load.
- Repeat the same transfer as many times as needed for the intended portion. Do not return the tongs to the rest part way through a topping.
- Once the intended portion has been added, carry the tongs back to the cutting board, lower them until stable, and release.

Do not skip a topping bowl, and do not go back to an earlier topping bowl once you have moved on.

Check after each topping bowl: the intended portion is in the mixing bowl and the tongs are back on the cutting board before the next topping bowl starts.

Expected state: the mixing bowl holds the intended portions of greens and all five toppings, undressed and untossed. The ingredient containers may still hold ingredients. The tongs and wooden spoon are on the cutting board.

Step 3: Dress the salad

Goal: dressing from both bottles is on the salad and both bottles are upright on their spots.

Use the two bottles one at a time, in either order. Each bottle gets one pour.

- The **right gripper** takes one bottle around the body and lifts it straight up off its spot.
- Carry it level to the mixing bowl and bring the spout over the middle of the mixing bowl, inside the rim.
- Tip the bottle to pour and squeeze steadily for a count of 3 seconds, then stop squeezing and bring the bottle upright again.
- Lift the bottle clear of the rim while it is upright, carry it back to its spot, lower it until it is standing on the tabletop, and release.
- The right gripper then takes the other bottle and repeats the same pour over the middle of the mixing bowl.
- Bring it upright, return it to its spot, lower it until it is standing on the tabletop, and release.

One pour from each bottle. Pour once from one bottle and return it, then pour once from the other. Do not pour a second time from either bottle, shake a bottle, or tap it on the rim.

Check: both bottles stand upright on their spots, dressing from both has landed on the salad rather than on the tabletop or down the outside of the mixing bowl, and the mixing bowl is still on its ring.

Step 4: Toss the salad

Goal: the dressing and the toppings are worked through the greens and the wooden spoon is back on the cutting board.

- The **right gripper** takes the wooden spoon from the cutting board and lifts it clear.
- Bring the wooden spoon down into the mixing bowl at the far side and push it to the bottom of the mixing bowl.
- Lift the wooden spoon up and over toward the middle of the mixing bowl, turning what is at the bottom over onto the top.
- Lower it back down at the next side and repeat. Make four lift and turn passes in all.
- Keep the wooden spoon inside the mixing bowl for the whole toss. Lift it clear only when the four passes are done.

Lift and turn, do not stir. Do not drag the wooden spoon in circles around the mixing bowl, do not press the salad down, and do not scrape the sides.

- The right gripper carries the wooden spoon back to the cutting board, lowers it until stable, and releases.

Check: the toppings are spread through the greens rather than sitting in a layer on top, nothing was thrown out of the mixing bowl during a pass, the mixing bowl is still on its ring, and the wooden spoon is back on the cutting board.

Step 5: Portion onto the two plates

Goal: both plates hold a portion and the mixing bowl is left near empty.

Work one plate at a time, taking the **far plate first**.

5.1 Take the tongs and bring the far plate

- **IF Config L:** the **left gripper** takes the tongs from the cutting board and lifts them clear. **IF Config M or R:** the **right gripper** takes them.
- The plate-side gripper takes the far plate by the rim, keeps it level, and brings it to the portioning position beside the mixing bowl, on the plate side.
- Lower the plate until it is stable on the tabletop, then release it.

5.2 Portion the far plate

- Bring the tongs down into the mixing bowl and close them on a portion.
- Move enough tong loads to place about half of what is in the mixing bowl onto the **far plate**, judged by eye.
- For each load, lift the portion straight up and clear of the rim, carry it level to the far plate, and lower it until it is just above the plate.
- Open the tongs to release each load onto the middle of the plate, then lift them clear.
- Once the tongs are clear, the plate-side gripper takes the filled far plate by the rim and returns it to its starting position. Lower it until stable, then release.

5.3 Bring and portion the near plate

- The plate-side gripper takes the near plate by the rim, keeps it level, and brings it to the portioning position beside the mixing bowl. Lower the plate until stable, then release.
- Go back to the mixing bowl and move what remains with the tongs the same way.
- Carry each load level to the **near plate** and release it onto the middle of the plate.
- Leave whatever will not come up in the tongs in the mixing bowl. Do not scrape the mixing bowl out.
- Once the tongs are clear, the plate-side gripper takes the filled near plate by the rim and returns it to its starting position. Lower it until stable, then release.

5.4 Return the tongs

- The ingredient-side gripper carries the tongs back to the cutting board, lowers them until stable, and releases.

Check: both plates hold a portion, the two portions look about the same size by eye, nothing overhangs a plate rim, both plates are back in their starting positions, the mixing bowl is near empty, and the tongs are back on the cutting board.

Step 6: End the episode

- Confirm all six ingredients were added, both bottles stand upright on their spots, the mixing bowl is near empty on its ring, and the wooden spoon and tongs are back on the cutting board.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Clear both plates and set the salad aside for disposal.
2. Empty the mixing bowl, wash and dry it, and set it back on its ring in the middle of the tabletop.
3. Wash and dry the tongs and wooden spoon and return them to the cutting board.
4. Wipe both bottle spouts clean, top up either bottle that no longer holds enough for a full pour, and stand both upright on their spots.
5. Refill the greens tub with one episode's worth of loose leaves, filled to below the rim.
6. Refill all five topping bowls, one topping each, none above the rim. Vary which topping goes in which topping bowl between episodes rather than rebuilding the same tray every time.
7. Replace any ingredient that is wilted, dried out, or clumped together.
8. Wipe the tabletop, set the six containers in the start zone for the next episode's config — the left side (Config L), the back-centre (Config M), or the right side (Config R) — and put both plates back empty on the opposite side, one above the other and not touching.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Note on the start position: the violations below were written for Config R (ingredient containers on the right, plates on the left, the right gripper on the tongs). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the ingredient containers are not in the start zone for the config, or the plates are not on the opposite side; a gripper reaches across the table for an ingredient container or a plate; the wrong gripper for the config holds the tongs or moves a plate; or the wrong IF line is followed.
- **SOP rule broken:** the start position, the plate side, and the same-side rule (the ingredient-side gripper takes the tongs, loads, and plates — the left gripper in Config L, the right gripper in Config M and R; the plate-side gripper brings and returns each plate; no arm reaches across the table; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the ingredient containers are before the first reach, then follow that config's IF lines through Steps 1, 2, and 5.

Violation: Filled out of order

- **Visible cue:** a topping goes in before the greens, or the topping bowls are worked in any order other than 1, 2, 3, 4, 5, such as reaching for the nearest full one or skipping ahead and coming back.
- **SOP rule broken:** Steps 1 and 2, tray order, not reach order: greens first, then topping bowl 1 through 5 in that order.
- **Coaching note:** greens first, then straight down the row.

Violation: Ingredient omitted

- **Visible cue:** no portion is taken from the greens tub or one of the five topping bowls.
- **SOP rule broken:** Steps 1 and 2 (add an intended portion from each of the six ingredient containers; the containers do not have to be emptied).
- **Coaching note:** confirm that all six ingredients were added before dressing the salad.

Violation: Two tools held at once

- **Visible cue:** a gripper picks up either bottle or the wooden spoon while the other gripper still holds the tongs, or the tongs are taken while another tool is still held.

- **SOP rule broken:** Steps 1 to 5, a tool goes back to the cutting board before the next tool is picked up, and never hold two tools at once.
- **Coaching note:** put it back before you pick the next one up.

Violation: Tool laid down outside its rest

- **Visible cue:** the tongs, either bottle, or wooden spoon is set on the tabletop, on the tray, across the bowl rim, or on a plate instead of going back to the cutting board or, for a bottle, its marked spot.
- **SOP rule broken:** Steps 1, 2, 3, 4, and 5.4, every utensil goes back to the cutting board between jobs and each bottle goes back to its marked spot.
- **Coaching note:** every tool has one home. It goes there every time.

Violation: Wrong arm on a tool or a zone

- **Visible cue:** the left gripper takes the tongs, either bottle, or wooden spoon or picks something off the tray, or the right gripper moves a plate.
- **SOP rule broken:** Steps 1 to 5, the right gripper owns the tray and all three tools, and the left gripper owns the plates.
- **Coaching note:** the right gripper loads, dresses, tosses, and portions. The left gripper brings and returns each plate.

Violation: Tool passed hand to hand

- **Visible cue:** the tongs, either bottle, or wooden spoon is transferred directly from one gripper to the other in mid air instead of going back to the cutting board first.
- **SOP rule broken:** Steps 1 to 5, nothing is passed hand to hand, and a tool that changes arms does so by going back to the cutting board first.
- **Coaching note:** the cutting board is the handoff. Nothing changes hands in the air.

Violation: More than one ingredient taken in a trip

- **Visible cue:** the tongs take from two containers before releasing over the mixing bowl, or two toppings are carried across in the same load.
- **SOP rule broken:** Steps 1 and 2, one ingredient at a time, never take from two containers on one trip and never combine two toppings in the tongs.
- **Coaching note:** one container per trip, every trip.

Violation: Ingredient gripped directly

- **Visible cue:** a bare gripper closes on leaves or a topping and moves it, or a gripper is used to push contents into the mixing bowl or off the tongs.
- **SOP rule broken:** Steps 1 to 5, never grip an ingredient directly, everything that goes in the mixing bowl or onto a plate gets there through the tongs.

- **Coaching note:** the tools touch the food. The grippers never do.

Violation: Load released from height

- **Visible cue:** the tongs open above the mixing bowl rim and the load falls the rest of the way, bouncing or scattering, instead of being lowered inside the rim first.
- **SOP rule broken:** Steps 1 and 2, bring the tongs down inside the rim of the mixing bowl before opening them.
- **Coaching note:** get inside the mixing bowl before you open.

Violation: Ingredient dragged over a container rim

- **Visible cue:** the tongs close above a container and scrape or drag the load over its side, or contents are pulled out over the rim rather than lifted straight up.
- **SOP rule broken:** Step 2, bring the tongs down inside the rim before closing them, then lift straight up.
- **Coaching note:** close inside, lift straight up.

Violation: Mixing bowl gripped, steadied, or moved

- **Visible cue:** a gripper closes on the mixing bowl or rests against it to hold it still, or the mixing bowl is lifted, slid, or turned at any point during the episode.
- **SOP rule broken:** Steps 1 to 5, never grip, steady, lift, or slide the mixing bowl, the ring holds it.
- **Coaching note:** the ring does that job. If the mixing bowl is moving, stop and reseal it.

Violation: Bottle carried or set down tipped

- **Visible cue:** either bottle is lifted away from the mixing bowl still tilted, dripping dressing on the tabletop, or is put back on its spot leaning or on its side.
- **SOP rule broken:** Step 3 (bring each bottle upright before lifting it clear of the rim, and lower each one until it is standing on its marked spot).
- **Coaching note:** upright before you move, upright when you let go.

Violation: Toss is a stir

- **Visible cue:** the wooden spoon is dragged in circles around the mixing bowl, pushed through the salad sideways, or used to press the salad down or scrape the sides, instead of going to the bottom and lifting up and over toward the middle.
- **SOP rule broken:** Step 4 (push the wooden spoon to the bottom and lift up and over toward the middle; lift and turn, do not stir, press, or scrape).
- **Coaching note:** go under it and turn it over. Never push it around.

Violation: Wrong number of toss passes

- **Visible cue:** the wooden spoon comes out of the mixing bowl after fewer than four lift and turn passes, or the tossing continues past four.
- **SOP rule broken:** Step 4 (make four lift and turn passes in all).
- **Coaching note:** count the passes out loud. Four, then out.

Violation: Wooden spoon lifted out mid toss

- **Visible cue:** the wooden spoon comes clear of the mixing bowl between passes.
- **SOP rule broken:** Step 4 (keep the wooden spoon inside the mixing bowl for the whole toss and lift it clear only when the four passes are done).
- **Coaching note:** the wooden spoon stays in until the count is done.

Violation: Wrong plating order

- **Visible cue:** the near plate is brought forward or filled before the far plate, or both portions go onto one plate.
- **SOP rule broken:** Step 5 (bring and fill one plate at a time, taking the far plate first).
- **Coaching note:** bring the far plate first and place one portion on each plate.

Violation: Plate handled incorrectly

- **Visible cue:** the left gripper touches the food-contact surface, tilts a plate, moves a plate while the tongs are over it, or releases a plate before it is stable.
- **SOP rule broken:** Step 5 (take each plate by the rim, keep it level, and move it only when the tongs are clear).
- **Coaching note:** grip the rim, keep the plate level, and wait for the tongs to clear.

Violation: Uneven portions

- **Visible cue:** the first scoop clearly takes most or nearly none of what is in the mixing bowl, leaving the two plates visibly different in size.
- **SOP rule broken:** Step 5.2 (take about half of what is in the mixing bowl on the first scoop, judged by eye).
- **Coaching note:** look at the mixing bowl before each tong load. Half, then the rest.

Violation: Bowl scraped out

- **Visible cue:** the tongs are dragged around the inside of the mixing bowl or tipped against its wall to work the last of the salad out after the near plate is filled.
- **SOP rule broken:** Step 5.3 (leave whatever will not come up in the tongs in the mixing bowl; do not scrape the mixing bowl out).
- **Coaching note:** what stays in the mixing bowl stays in the mixing bowl.

Violation: Portion dropped instead of lowered

- **Visible cue:** the tongs open above a plate and the portion falls the rest of the way, scattering across the plate or off its rim, instead of being lowered to just above the plate first.
- **SOP rule broken:** Step 5.2 and Step 5.3 (lower each portion until it is just above the plate, then open the tongs).
- **Coaching note:** take it all the way down before you open.

Violation: Salad pressed or packed

- **Visible cue:** the wooden spoon, tongs, or a gripper presses down on the contents of the mixing bowl or on a plated portion to flatten or shape it.
- **SOP rule broken:** Steps 4 and 5, never press or pack the salad down.
- **Coaching note:** leave it how it lands.

Violation: Gripper released over the wrong place

- **Visible cue:** a gripper opens over the tray, over a topping bowl, or over open tabletop, or a tool is released anywhere other than the cutting board or, for a bottle, its marked spot.
- **SOP rule broken:** Steps 1 to 5, a gripper opens only over the mixing bowl, over a plate, or at a stable placement on the cutting board.
- **Coaching note:** know where it lands before you open the gripper.

Violation: Spill collected mid episode

- **Visible cue:** an arm stops to pick up, sweep, or push leaves or toppings that have landed on the tabletop back into the mixing bowl or a container.
- **SOP rule broken:** Steps 1 to 5, spilled leaves or toppings on the tabletop are left where they fall.
- **Coaching note:** keep going. Spills are cleared in the reset, not on camera.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the wooden spoon or tongs off the cutting board, either bottle off its spot or lying down, one of the six ingredients omitted, or an arm away from home.
- **SOP rule broken:** Step 6 (confirm all six ingredients were added, both bottles are upright, the mixing bowl is near empty, and both utensils are on the cutting board; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).

- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective ingredient:** greens found wilted or frozen together, or a topping found clumped into a mass the tongs cannot separate. Replace it before the next episode.
- **Either dressing bottle runs empty or will not pour** under a steady squeeze, leaving the salad partly undressed. Top it up or clear the spout before the next episode.
- **Tool or rest fault:** tongs that will not close on a load, a wooden spoon that comes loose in the gripper, or a utensil that will not remain stable on the cutting board.
- **Mixing bowl leaves its ring** under its own weight, or the ring slides on the tabletop, so the mixing bowl cannot be worked without being held. Reseat both before the next episode.
- **A plate or the tray shifts out of its zone** during the episode.

Annotation subtasks (from SOP)

1. Take the tongs from the rest
2. Move the intended portion of greens from the tub to the mixing bowl
3. Move the intended portion of a topping from its bowl to the mixing bowl
4. Return the tongs to the rest
5. Take one dressing bottle from its spot
6. Pour dressing from the bottle over the mixing bowl
7. Return the bottle to its spot
8. Take the wooden spoon from the cutting board
9. Toss the salad with four lift and turn passes
10. Return the wooden spoon to the cutting board
11. Bring one plate to the portioning position with the plate-side gripper
12. Move one portion from the mixing bowl to the plate with the tongs
13. Return the filled plate to its starting position with the plate-side gripper
14. Return both arms home and end the episode